Task 1:

1. Identify the Needs

1.1-Voltage and Current Requirements for Each Load:

1-Load A (Relay):

- **Operating Voltage:** 9V
- **Operating Current:** 50 mA

6. COIL DATA CHART (AT20°C)

Coil Sensitivity	Coil Voltage Code	Nominal Voltage (VDC)	Nominal Current (mA)	Coil Resistance (Ω) ±10%	Power Consumption (W)	Pull-In Voltage (VDC)	Drop-Out Voltage (VDC)	Max-Allowable Voltage (VDC)
SRD (High Sensitivity)	03	03	120	25	abt. 0.36W	75%Max.	10% Min.	120%
	05	05	71.4	70				
	06	06	60	100				
	09	09	40	225				
	12	12	30	400				
	24	24	15	1600				
	48	48	7.5	6400				
SRD (Standard)	03	03	150	20	abt. 0.45W	75% Max.	10% Min.	110%
	05	05	89.3	55				
	06	06	75	80				
	09	09	50	180				
	12	12	37.5	320				
	24	24	18.7	1280				
	48	48	10	4500	abt. 0.51W			

2-Load B (HC-SR04 Ultrasonic Sensor):

- **Operating Voltage:** 5V
- **Operating Current:** 15mA

Electric Parameter

Working Voltage	**DC 5 V**
Working Current	**15mA**

1.2-Battery Voltage Analysis:

- The fully charged 3S Li-ion battery supplies 12.6V, which is higher than the operational voltage required by both Load A (9V) and Load B (5V). Consequently, step-down voltage regulation is required for both outputs.

1.3-Reverse Polarity Protection:

- **Yes**, reverse polarity protection is essential to safeguard the voltage regulators and sensitive downstream ICs from permanent damage in the event of accidental reverse battery connection. A Schottky diode (MBRS130L) was implemented at the primary input to provide low forward voltage drop and efficient protection.

2. Choose the Components

2.1-Regulator Selection:

1-For Load B (5V - HC-SR04 Sensor):

- **Selected Regulator:** Buck Converter (LTC3630)
- **Reason:** Stepping down from 12.6V to 5V is a large voltage difference. A Buck converter was chosen because it is much more efficient than a linear regulator and doesn't waste excess power as heat, which helps save battery life.

2-For Load A (9V - Relay):

- **Selected Regulator:** LDO / Linear Regulator (LT3080)
- **Reason:** Stepping down from 12.6V to 9V is a small voltage difference (3.6V). An LDO was selected because it is simple to implement, provides a clean DC output, and handles the low current demand (50mA) with minor power loss.

2.2-Protection Circuit Selection & Justification:

1-Reverse Polarity Protection:

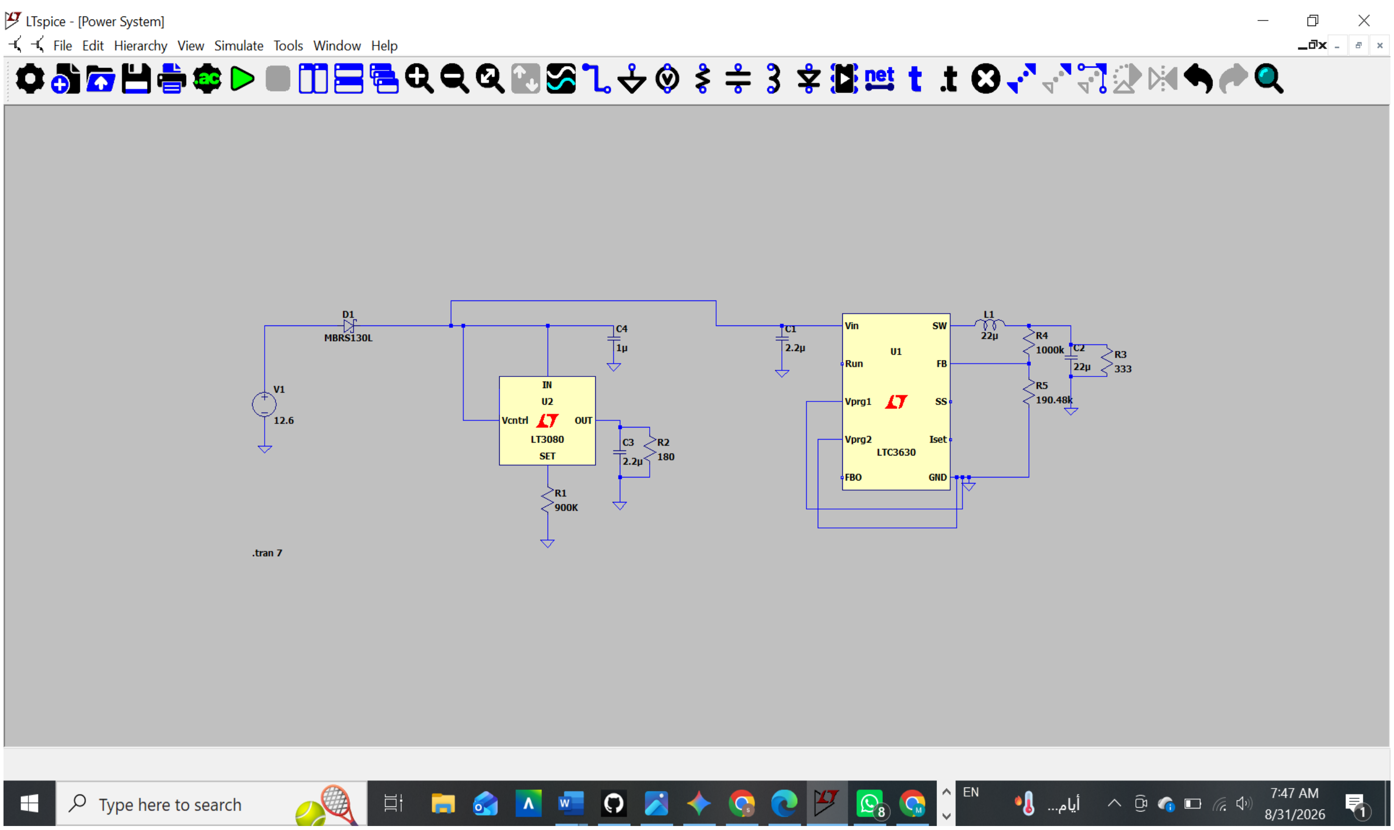
- An MBRS130L Schottky diode was added in series at the battery input. It protects the circuit if the battery is connected backwards. A Schottky diode was picked because it has a very low forward voltage drop ($\approx$ 0.3V - 0.4V), saving energy.

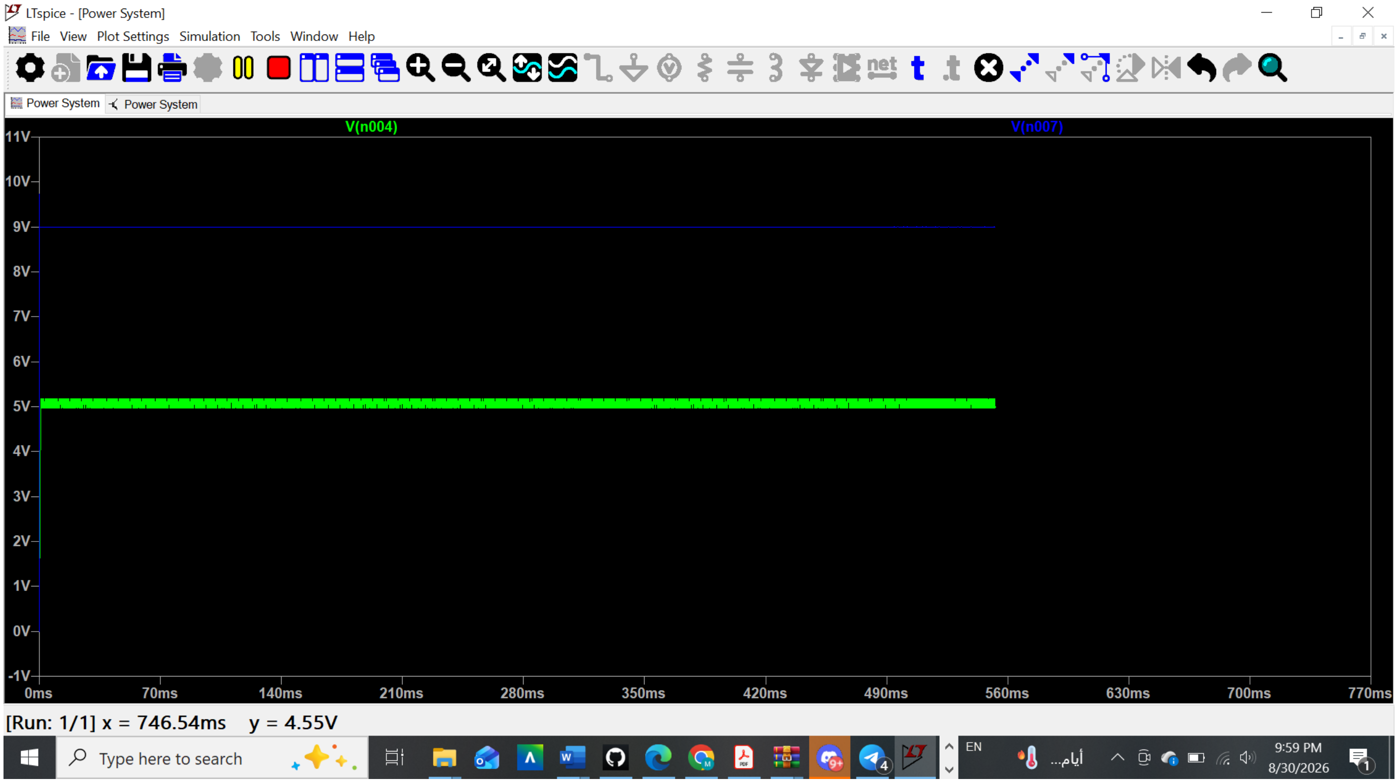
2-Relay Protection (Practical Note):

- In this simulation, the relay was modeled as a simple resistor (180 Ohm). However, in a real hardware build, a Flyback Diode must be placed in parallel across the relay coil to safely bleed off voltage spikes (Back-EMF) when the relay turns off.

3-Check the Datasheet

- **Schottky Diode (MBRS130L - Reverse Polarity Protection):**
  - **Datasheet Limits:** Reverse Voltage $V_R = 30V$, Forward Current $I_{F(AV)} = 1A$.
  - **Justification:** The 30V rating safely handles the 12.6 battery peak. It protects the entire circuit against reversed battery connections while maintaining a low forward voltage drop $\approx 0.3V$, minimizing power loss.
- **Linear Regulator (LT3080 - 9V Output for Relay):**
  - **Datasheet Limits:** Input Voltage $V_{IN} = 1.2V$ to $36V$, Max Output Current $I_{OUT} = 1.1A$.
  - **Justification:** Its input range easily covers the 12.6V Li-ion battery.Because the voltage differential 12.6V→9V and relay current draw are small, power dissipation remains minimal while providing clean, ripple-free voltage.
- **Buck Regulator (LTC3630 - 5V Output for Ultrasonic Sensor):**
  - **Datasheet Limits:** Input Voltage $V_{IN} = 4V$ to $65V$, Output Current $I_{OUT} = 500MA$.
  - **Justification:** The wide input range safely accepts the 12.6V input. A switching buck topology is chosen over an LDO to significantly boost efficiency (up to ≈88%) during the 12.6V→ 5V step-down, preventing unnecessary battery drain and excess heat generation.





Task 2:Component Functions & Purpose

- **Capacitors (C1)**
  - **Functions:** Filtering high-frequency noise, smoothing voltage ripple, decoupling power rails, and storing energy temporarily.
  - **Why needed:** Placed at input/output terminals to stabilize voltage, prevent sudden drops during high load bursts, and filter out switching noise generated by DC-DC regulators.
- **Resistors (R11)**
  - **Functions:** Current limiting, voltage division, pull-up/pull-down settings, and feedback networks.
  - **Why needed:** Set the feedback loop (V_{FB}) to define output voltage levels, set soft-start times, limit currents entering IC pins, and pull lines to safe logic levels.
- **Inductors (L1)**
  - **Functions:** Energy storage in a magnetic field, current smoothing, and filtering high-frequency noise.
  - **Why needed:** Crucial for buck switching regulators to store energy during the ON cycle and release it during the OFF cycle, converting pulsed signals into steady DC current.
- **Voltage Regulators (U1)**
  - **Functions:** Step-down voltage conversion (Buck) and line/load regulation.

    - **Why needed:** Safely drop the high battery voltage (12.6V) down to stable low rails (5V, 9V, or 10V) needed by sensors and microcontrollers without damage.

- **Diodes (D1)**
    - **Functions:** Unidirectional current flow, reverse polarity protection, and flyback/freewheeling protection.
    - **Why needed:** Prevent current from flowing backward if the battery is connected incorrectly, and provide a freewheeling path for switching inductors to dissipate voltage spikes safely.
- **Connectors (Terminals, Pins)**
    - **Functions:** Physical interfacing, mechanical securing, and electrical current routing.
    - **Why needed:** Provide reliable plug-and-play connections between the battery supply and the PCB, as well as easy wiring to external loads (relays, sensors).